

## The Optimization Problem

Let  $C = \{1, \dots, n\}$  denote a set of clients and  $J = \{1, \dots, m\}$  a set of candidate microhub locations. Each microhub  $j \in J$  is characterized by a fixed opening cost  $f_j \geq 0$  measured in monetary units, a maximum service capacity  $L_j > 0$  measured in units of client demand, a minimum operational utilization threshold  $\mu_j \geq 0$ , and an operational cost per unit of demand served  $o_j \geq 0$ . Each client  $c \in C$  generates a deterministic demand  $q_c > 0$  measured in the same units as hub capacity, incurs a walking distance  $d_{cj} \geq 0$  measured in minutes when assigned to microhub  $j$ , and is associated with a service priority weight  $w_c > 0$  that reflects the relative importance of rapid access. A client is considered covered by microhub  $j$  if  $d_{cj} \leq D_{\max}$ , where  $D_{\max} > 0$  is a system-wide maximum walking-distance threshold representing the operator's service quality standard.

Let  $y_j \in \{0, 1\}$  indicate whether microhub  $j$  is opened, and let  $x_{cj} \in \{0, 1\}$  indicate whether client  $c$  is assigned to microhub  $j$ . The MLPAP minimizes a weighted combination of two cost components: the total facility cost, which combines fixed opening and demand-proportional operational costs, and the total weighted assignment cost, which penalizes walking distance in proportion to client priority:

$$\min_{\mathbf{x}, \mathbf{y}} \underbrace{\sum_{j \in J} (f_j y_j + o_j \sum_{c \in C} q_c x_{cj})}_{\text{facility cost}} + \underbrace{\sum_{c \in C} \sum_{j \in J} w_c d_{cj} x_{cj}}_{\text{weighted assignment cost}} \quad (1)$$

The facility cost component captures both the capital expenditure associated with opening a microhub and the variable cost of processing the demand routed through it. The weighted assignment cost penalizes long walking distances in proportion to client priority, thereby favoring nearby access for high-priority users and reflecting the service-oriented nature of pedestrian-access delivery systems. In addition to conventional capacity and assignment requirements, the model explicitly incorporates a minimum operational utilization threshold for each active hub. This condition reflects the practical requirement that microhubs must maintain a sufficient level of throughput to justify staffing, storage infrastructure, and operational viability within dense urban environments [14, 16, 47]. The problem is subject to the following constraint sets:

$$\sum_{j \in J} x_{cj} = 1, \quad \forall c \in C, \quad (2)$$

$$\mu_j y_j \leq \sum_{c \in C} q_c x_{cj} \leq L_j y_j, \quad \forall j \in J, \quad (3)$$

$$x_{cj} \leq y_j, \quad \forall c \in C, j \in J, \quad (4)$$

$$d_{cj} x_{cj} \leq D_{\max}, \quad \forall c \in C, j \in J, \quad (5)$$

$$P_{\min} \leq \sum_{j \in J} y_j \leq P, \quad (6)$$

$$x_{cj}, y_j \in \{0, 1\}, \quad \forall c \in C, j \in J, \quad (7)$$

where  $P \in \mathbb{Z}^+$  is a budget ceiling on the total number of open microhubs and  $P_{\min} \in \mathbb{Z}^+$  is a minimum territorial coverage requirement. Equation (2) enforces single assignment, ensuring that each client is allocated to exactly one microhub. Equation (3) imposes both minimum-utilization and maximum-capacity requirements, thereby defining lower and upper throughput limits for every active facility. Equation (4) establishes hub-activation coherence by preventing clients from being assigned to closed microhubs. Equation (5) captures pedestrian accessibility by limiting assignments to facilities located within the maximum admissible walking distance. Equation (6) controls the number of active hubs through lower and upper bounds on facility activation. Finally, Equation (7) defines the binary domain of the decision variables.

The decision vector is  $\mathbf{z} = (\mathbf{y}, \mathbf{x}) \in \{0, 1\}^m \times \{0, 1\}^{nm}$ , with dimensionality  $\mathcal{O}(nm)$ . The MLPAP is closely related

to the Capacitated Facility Location Problem (CFLP) [17, 19], but extends the classical formulation by simultaneously incorporating pedestrian-access coverage, minimum operational utilization, and hub-budget constraints within a unified binary structure. The decision version remains NP-hard [22], while the combination of assignment, coverage, utilization, and cardinality restrictions substantially tightens the feasible region relative to conventional facility-location formulations.

The interaction among Equations (3), (5), and (6) produces a highly restrictive and irregular feasible space. Equation (3) prevents the activation of underused hubs that would otherwise reduce assignment cost, while Equation (5) rules out assignments beyond a prescribed accessibility threshold. At the same time, Equation (6) globally couples all hub-opening decisions through a cardinality restriction imposed on the binary vector  $\mathbf{y}$ . Together, these mechanisms generate a multimodal combinatorial landscape in which small modifications to the hub-activation pattern can simultaneously affect coverage feasibility, assignment structure, and capacity balance across multiple facilities.

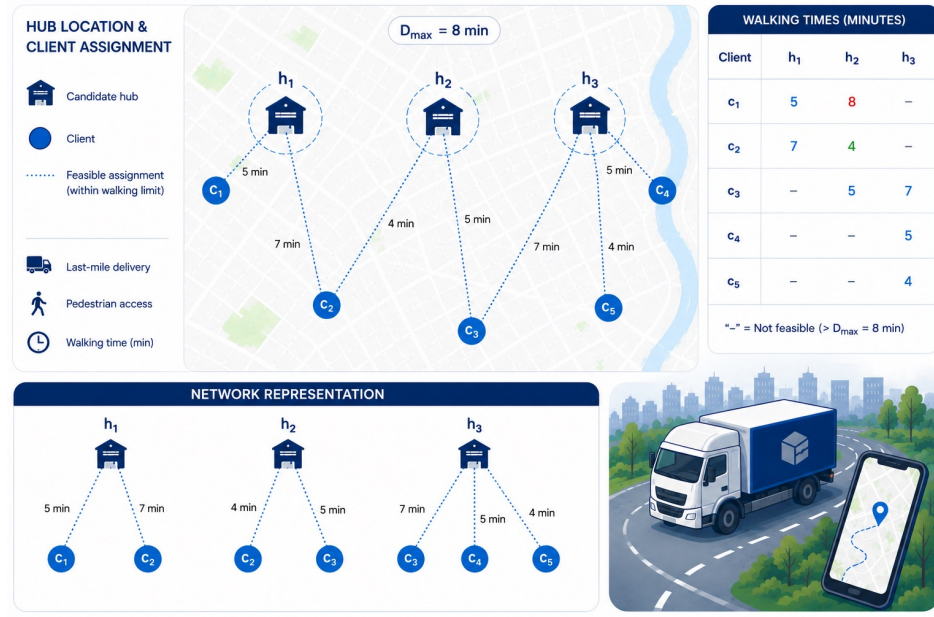
The treatment of infeasible solutions adopted in this work follows a strict penalization strategy without repair heuristics. Specifically, for any candidate solution  $\mathbf{z}$  that violates one or more constraints, the evaluated objective is augmented by an additive penalty proportional to the total magnitude of violation:

$$\tilde{f}(\mathbf{z}) = f(\mathbf{z}) + \pi \cdot v(\mathbf{z}), \quad (8)$$

where  $f(\mathbf{z})$  is the raw objective value of Equation (1),  $v(\mathbf{z})$  represents the aggregated magnitude of constraint violations across Equations (2)–(6), and  $\pi$  is a fixed penalty coefficient, which controls the relative importance assigned to feasibility with respect to the raw objective value.

This rejection-based penalization scheme was deliberately selected to preserve methodological consistency across all compared algorithms. In contrast to repair-based approaches, which often embed problem-specific heuristics inside the search process [64, 65], the adopted strategy maintains the original search dynamics of each metaheuristic while providing a unified feasibility-handling mechanism.

To clarify the structure of the MLPAP, consider a small instance with five clients,  $C = \{c_1, c_2, c_3, c_4, c_5\}$ , and three candidate microhub locations,  $H = \{h_1, h_2, h_3\}$ . The instance is intentionally simple so that the objective value and all constraints can be verified manually. The maximum admissible walking distance is  $D_{\max} = 8$  minutes, the maximum number of open hubs is  $P_{\max} = 2$ , and the minimum number of open hubs is  $P_{\min} = 1$ . Figure 1 illustrates the spatial structure of the instance before solving the optimization problem. It shows the candidate hubs, the clients, and the feasible client–hub walking connections defined by the threshold  $D_{\max}$ . This representation highlights that the problem is not only to select facilities, but also to determine which feasible assignments should be activated under capacity, accessibility, and hub-budget restrictions. The parameters and walking times are summarized in Table 1.



**Figure 1.** Spatial representation of the MLPAP example instance. The figure shows five clients, three candidate microhubs, and all feasible client–hub walking connections within the maximum admissible walking distance  $D_{\max} = 8$  minutes.

**Table 1.** Description of the MLPAP example instance.

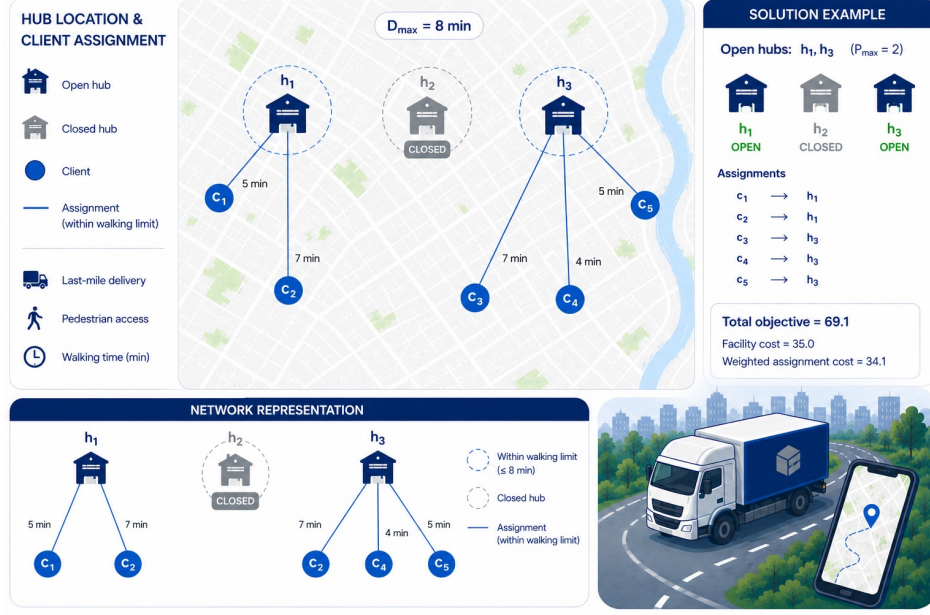
| Candidate microhubs |                 |       |       |         |       |  |
|---------------------|-----------------|-------|-------|---------|-------|--|
| Hub $h_j$           | Type            | $f_j$ | $L_j$ | $\mu_j$ | $o_j$ |  |
| $h_1$               | Candidate hub 1 | 20    | 7     | 4       | 0     |  |
| $h_2$               | Candidate hub 2 | 25    | 8     | 4       | 0     |  |
| $h_3$               | Candidate hub 3 | 15    | 6     | 4       | 0     |  |

| Clients and walking times $d_{cj}$ |                              |       |       |       |                 |  |
|------------------------------------|------------------------------|-------|-------|-------|-----------------|--|
| Client $c$                         | Demand and priority          | $h_1$ | $h_2$ | $h_3$ | Feasible hubs   |  |
| $c_1$                              | $q_{c_1} = 2, w_{c_1} = 1.0$ | 5     | 8     | 11    | $h_1, h_2$      |  |
| $c_2$                              | $q_{c_2} = 3, w_{c_2} = 1.5$ | 7     | 4     | 10    | $h_1, h_2$      |  |
| $c_3$                              | $q_{c_3} = 3, w_{c_3} = 1.2$ | 9     | 5     | 7     | $h_2, h_3$      |  |
| $c_4$                              | $q_{c_4} = 2, w_{c_4} = 1.0$ | 8     | 9     | 5     | $h_1, h_3$      |  |
| $c_5$                              | $q_{c_5} = 1, w_{c_5} = 1.3$ | 6     | 7     | 4     | $h_1, h_2, h_3$ |  |

A feasible solution is obtained by opening hubs  $h_1$  and  $h_3$ , that is,  $\mathbf{y} = (1, 0, 1)$ . The associated assignment is:  $c_1 \rightarrow h_1$ ,  $c_2 \rightarrow h_1$ ,  $c_3 \rightarrow h_3$ ,  $c_4 \rightarrow h_3$ , and  $c_5 \rightarrow h_3$ . Figure 2 presents this solved configuration. The closed hub  $h_2$  is excluded from the assignment structure, while all selected client–hub links satisfy the walking-distance threshold. The figure also summarizes the objective value and its decomposition into facility and weighted assignment costs.

In binary terms, this means that  $x_{c_1 h_1} = 1$ ,  $x_{c_2 h_1} = 1$ ,  $x_{c_3 h_3} = 1$ ,  $x_{c_4 h_3} = 1$ , and  $x_{c_5 h_3} = 1$ , while all remaining assignment variables are equal to zero.



**Figure 2.** Solved configuration of the MLPAP example. Hubs  $h_1$  and  $h_3$  are opened, hub  $h_2$  remains closed, and each client is assigned to exactly one open hub within the walking-distance threshold.

The facility cost is given by the sum of the opening costs of the active hubs:

$$f_1 y_1 + f_3 y_3 = 20 + 15 = 35.$$

Since  $o_j = 0$  for all hubs in this example, there is no additional demand-proportional operating cost. The weighted assignment cost is:

$$1.0 \cdot 5 + 1.5 \cdot 7 + 1.2 \cdot 7 + 1.0 \cdot 5 + 1.3 \cdot 4 = 5 + 10.5 + 8.4 + 5 + 5.2 = 34.1.$$

Therefore, the total objective value is:

$$f(\mathbf{z}) = 35 + 34.1 = 69.1.$$

The solution satisfies all constraints of the model. Equation (2) is satisfied because every client is assigned to exactly one hub. Equation (3) is also satisfied: hub  $h_1$  serves demand  $q_{c_1} + q_{c_2} = 2 + 3 = 5$ , which lies within the interval  $[\mu_1, L_1] = [4, 7]$ , and hub  $h_3$  serves demand  $q_{c_3} + q_{c_4} + q_{c_5} = 3 + 2 + 1 = 6$ , which lies within  $[\mu_3, L_3] = [4, 6]$ . Hub  $h_2$  remains closed and serves no demand, consistently with  $y_2 = 0$ . Equation (4) is satisfied because no client is assigned to a closed hub. Equation (5) is satisfied because all selected walking times are below or equal to  $D_{\max} = 8$  minutes. Finally, Equation (6) is satisfied because exactly two hubs are opened, and therefore  $P_{\min} \leq \sum_{j \in J} y_j \leq P_{\max}$  holds with  $1 \leq 2 \leq 2$ .

This example illustrates the role of the minimum-utilization requirement. Opening an additional facility may reduce walking distances, but it can also create an infeasible solution if the demand assigned to that facility is insufficient to satisfy its minimum utilization threshold. Hence, the model does not only seek nearby assignments; it also enforces operationally meaningful hub activation.